

Next-Day Municipal Service-Demand Forecasting Across Four Cities: A Benchmark with a Controlled Simulated Capacity-Allocation Evaluation

Anonymous Submission

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Abstract

City agencies use short-horizon forecasts of 311 service-request volume as a planning input, but forecast accuracy is only a proxy for the decisions a forecast supports. We build a reproducible benchmark from 36.8 million acquired service-request records (30.2 million retained after documented exclusions) across four U.S. cities (New York, Chicago, San Francisco, Austin; 2020–2025), harmonized into documented service families and joined with NOAA station weather, and we evaluate next-day forecasters jointly with a transparent, explicitly simulated family-level capacity-allocation model under pre-specified hypothetical service-pressure regimes, with capacity budgets frozen from training data only and all model selection performed on validation data only. Calendar augmentation reduces next-day error in every city (8.6–14.0%, all 28-day block-bootstrap intervals excluding zero); target-day weather helps three cities and adds nothing in the fourth, a retained null. Pooled cross-city training, temporally censored so no training row is contemporaneous with any evaluation window, raises error significantly in three cities; zero-shot transfer degrades error by up to 149%. In the simulation, feeding point forecasts into an expected-value optimizer is a failure mode — the marginal-value profile a degenerate forecast induces is piecewise flat, so the optimizer loses even to naive proportional apportionment in all 12 city-regime settings — while the full predictive distribution of a single quantile model restores an allocation gradient and beats its own median and implied-mean representations in all 12 settings; against that distribution-aware optimizer, proportional apportionment remains competitive to slightly better (11 of 12, by 0.1–6.5%, one reversal), conditional on the stylized carryover dynamics and the neutral equal-weights objective. Forecast quality transfers to simulated decision value within a fixed policy in all 12 settings, and validation-time selection by decision loss mostly coincides with selection by accuracy (9 of 12), occasionally gaining and never harming. All data are public, every number regenerates from one command under a guard suite that enforces the protocol, and the simulation makes no claim about any city’s actual operations.

1 Introduction

Municipal 311 systems receive millions of service requests each year — heating failures, water leaks, street damage, sanitation problems, noise — and agencies plan finite daily service capacity against this demand. Short-horizon forecasts of request volume are a natural planning input, and a growing literature shows that public signals such as weather and calendar structure improve such forecasts. But planners do not consume mean absolute error; they consume allocations, and the mapping from predictive accuracy to allocation quality is neither linear nor guaranteed.

This paper contributes, first, a reproducible four-city benchmark of next-day reported 311 demand — New York, Chicago, San Francisco, and Austin, 2020–2025, built entirely from official open-data APIs with versioned harmonization rules and provenance manifests — and, second, a controlled, explicitly simulated evaluation of what forecast differences are worth to a family-level

capacity allocation under pre-specified hypothetical service-pressure regimes. The governing question is:

Across heterogeneous municipal 311 reporting systems, under pre-specified hypothetical service-pressure regimes, when do differences in next-day forecasts of reported administrative demand produce materially different outcomes in a transparent simulated family-level capacity allocation, and when are predictive rankings sufficient proxies for simulated decision rankings?

Four design commitments distinguish the evaluation protocol. **(i) Frozen environments:** capacity budgets derive from training data only and are identical at model-selection time and at the single test evaluation. **(ii) Validation-only selection:** no headline configuration is chosen using test outcomes. **(iii) Temporal censoring:** pooled and transfer models never train on rows contemporaneous with any evaluation window. **(iv) Controlled uncertainty contrast:** the value of distributional information is measured within one fitted quantile model — its median, implied-mean, and full-distribution representations — holding everything else fixed. A guard suite of automated tests enforces each commitment and fails the build if any is violated.

Throughout, we are explicit about what the data can and cannot support. Reported requests are not social need (Kontokosta and Hong, 2021); the allocation layer is a simulation over abstract request-equivalent capacity units whose parameters are varied over a pre-specified grid rather than asserted; and no causal, deployment, or operational-guidance claim is made. The study is an empirical bridge between the forecasting literature, which stops at accuracy, and the decision-focused learning literature (Elmachtoub and Grigas, 2022; Mandi et al., 2024), which rarely touches public-sector data.

2 Related Work

Public signals in operational forecasting. Cui et al. (2018) established, in retail, that publicly available signals (social media) measurably improve daily operational forecasts; Steinker et al. (2017) did the same for weather in e-commerce fulfilment. An earlier single-city study by the author brought this template to municipal service demand on NYC 311. The present paper tests whether that finding is a property of New York or of 311 systems generally, and then asks what the gains are worth inside a controlled simulated allocation.

311 analytics. 311 data underpin a largely single-city, prediction-centric literature: Xu et al. (2017) forecast Chicago 311 demand in space and time with an adaptive kernel and motivate the exercise by resource-allocation relevance without evaluating an allocation; Cheng et al. (2022) predict request volumes and their correlates in Miami-Dade County; Kontokosta et al. (2017) and Kontokosta and Hong (2021) document socio-spatial disparities in the propensity to report, establishing that 311 volumes measure *reported* demand rather than need. We inherit the latter caveat explicitly. To our knowledge no prior work evaluates 311 forecasting across multiple cities under one protocol, nor couples it to an explicit allocation loss with frozen pre-test environments.

From prediction to decisions. That predictive accuracy and decision quality are distinct objectives is the founding observation of prescriptive analytics (Bertsimas and Kallus, 2020) and decision-focused learning: the SPO framework trains models against the downstream objective (Elmachtoub and Grigas, 2022), task-based end-to-end learning does so in stochastic programs (Donti et al., 2017), and Mandi et al. (2024) survey and benchmark the field. Those benchmarks are

dominated by synthetic costs or energy/inventory/routing settings. We contribute complementary public-sector evidence: rather than proposing a training method, we measure empirically where accuracy-optimized models, distributional information, and decision-aware validation selection do and do not change simulated allocation outcomes.

Capacity planning under time-varying demand. Setting service capacity against predictable daily variation is a classical operations problem (Green et al., 2007); our simulation is a deliberately transparent discrete-time instance with integer capacity units, family-level objectives, and carryover of unresolved request-equivalents, designed for auditability rather than queueing fidelity.

Cross-city learning and urban computing. Urban computing treats the city as a sensing-and-decision platform (Zheng et al., 2014); cross-city transfer has been studied mainly for traffic and crowd-flow prediction (Wang et al., 2019), and pooled training across related series is standard in modern forecasting (Salinas et al., 2020). We test pooled-versus-local training and zero-shot transfer under explicit temporal censoring, and carry the comparison through to simulated decision quality.

3 Problem Formulation

Prediction task. Let $y_{c,s,t}$ denote the number of 311 requests created in city c and harmonized service family $s \in \mathcal{S}_c$ on calendar day t , where $\mathcal{S}_c$ is the city’s *active* family set (eight families in New York, San Francisco, and Austin; seven in Chicago, where residential noise is structurally absent from the intake taxonomy). At the end of day t (the information cutoff), a forecaster produces a point forecast $\hat{y}_{c,s,t+1}$ and, for the probabilistic model, quantile forecasts $\hat{q}_{c,s,t+1}^\tau$, $\tau \in \{.05, .25, .50, .75, .95\}$. Endogenous features use observations through day t only; calendar features of day $t+1$ are deterministic; target-day weather uses the realized observation for $t+1$ as a stand-in for a day-ahead weather forecast (assumption A3, bounded by a lagged-weather variant).

Simulated allocation. Each evening a simulated planner allocates a fixed budget of B_c abstract capacity units across active families, $x_{c,s,t+1} \in \mathbb{Z}_{\geq 0}$, $\sum_s x_{c,s,t+1} = B_c$; a unit resolves up to κ request-equivalents the next day. Unresolved workload carries over:

$$\begin{aligned} w_{c,s,t+1} &= b_{c,s,t} + y_{c,s,t+1}, & \text{resolved} &= \min(w_{c,s,t+1}, \kappa x_{c,s,t+1}), \\ u_{c,s,t+1} &= w_{c,s,t+1} - \text{resolved}, & b_{c,s,t+1} &= (1 - \alpha) u_{c,s,t+1}, \end{aligned} \quad (1)$$

where b is simulated unresolved request-equivalent carryover (observable state at decision time) and α an abandonment rate ($\alpha = 0$ primary). The decision loss over the test horizon $\mathcal{T}$ is

$$L_c = \sum_{t \in \mathcal{T}} \sum_{s \in \mathcal{S}_c} \omega_s u_{c,s,t}, \quad (2)$$

with EQUAL weights $\omega_s = 1$ as the primary, neutral objective (one illustrative normative-priority scenario is a labeled sensitivity). Because $u_{c,s,t}$ contains the carried stock, a request that remains unresolved for k days contributes k times to (2): the loss is a *waiting-time-weighted cumulative unresolved stock*, not a daily flow. Under the scarce regime the simulated system is structurally under-capacitated (supercritical), so this stock grows over the horizon and daily losses — and the daily loss *differentials* between policies — can carry a deterministic trend; Section 8 reports a trend diagnostic and the inference framing follows it. Hypothetical service-pressure regimes set κB_c to 70%, 90%, and 110% of the city’s TRAINING-window mean daily demand; the resulting integer

budgets are frozen before any model selection and used unchanged for validation-stage selection and the single test evaluation. Nothing in this construction estimates any city’s actual capacity, productivity, or queues.

Policies. All policies receive the same observable state and allocate the full budget: *uniform* (equal units; forecast-free floor); *proportional* (largest-remainder apportionment of $\hat{y}_{c,s,t+1} + b_{c,s,t}$); and *greedy expected-value*, which exactly maximizes $\sum_s \omega_s \mathbb{E}[\min(D_s, \kappa x_s)]$ over integer allocations, where D_s is the forecast demand-plus-carryover distribution — degenerate for point forecasts, piecewise-linear from the quantile model. Greedy is optimal for this objective because $\mathbb{E}[\min(D, c)]$ is concave in c . A hindsight myopic reference (greedy on realized demand) appears in supplementary diagnostics only; being per-day myopic it is not a horizon-optimal bound and is never used as a denominator or target.

Controlled uncertainty contrast. The value of distributional information is isolated within ONE fitted quantile model: a degenerate-median arm, an implied-mean arm, and a full-distribution arm differ only in how the same predictive distribution enters the greedy policy. The separately trained point models serve as forecasting benchmarks, not as the uncertainty comparator.

Evaluation questions. For each city and regime we compare forecast accuracy (MAE, with paired 28-day moving-block bootstrap intervals; pinball loss and coverage for quantiles) and simulated decision loss (2) (raw losses, absolute paired daily differences with block-bootstrap intervals, and percentage reduction relative to the uniform floor), asking when predictive and decision orderings agree, whether distributional information changes outcomes, and whether validation-time selection by decision loss differs from selection by accuracy.

4 Data

311 service requests. We use the official open-data portals of four U.S. cities, selected by pre-stated criteria (API uniformity, coverage of the 2020–2025 study window — set by the scope of NYC’s current dataset, decision D14 — category granularity, and heterogeneity in size and climate; see the feasibility audit in the companion repository): New York ([erm2-nwe9](#), NYC Open Data), Chicago ([v6vf-nfxy](#), current 311 system, live since December 2018), San Francisco ([vw6y-z8j6](#), DataSF, PDDL license), and Austin ([xwdj-i9he](#)). Acquisition uses server-side daily aggregation (counts per day $\times$ native category) through the SODA API; every query URL, retrieval timestamp, and SHA-256 checksum is recorded in versioned manifests. Chicago rows flagged as duplicates by the source system are removed (523,055 requests), as are documented non-service categories: Chicago’s “311 INFORMATION ONLY CALL” log entries (4,184,158 records) and aviation-routed aircraft-noise filings (1,918,574 records), and San Francisco’s administrative duplicate markers. The study window is 2020-01-01 through 2025-12-31, the full scope of NYC’s current dataset. As an acquisition-integrity check, the NYC layer was produced by two independent methods — server-side aggregation and client-side aggregation of a streamed export — which agreed exactly (270,978 aggregated rows).

Harmonized service families. Each city’s native taxonomy ($\sim$ one to three hundred categories) is mapped to eight service families — noise; housing/building; water/sewer; public safety; streets/transport; sanitation; parks/trees; other — by an ordered, case-insensitive, versioned keyword rule set applied identically to all cities. The rules were audited against each city’s real category

Table 1: Dataset summary after preprocessing (study window 2020-01-01 to 2025-12-31; eight defined service-family categories with city-specific active sets — Chicago has seven because the noise family is structurally absent; no synthetic data).

City	Requests kept	Native cat.	Panel rows	Days	Mean/day
Austin	1,487,588	346	17,536	2192	679
Chicago	4,683,510	107	15,344	2192	2,137
New York	19,669,371	271	17,536	2192	8,973
San Francisco	4,374,676	136	17,536	2192	1,996

inventory and frozen before any model was trained; after the freeze, the unmapped “other” share is 6.2% (New York), 6.9% (Chicago), 8.8% (San Francisco), and 23.5% (Austin, whose intake includes large animal-services and explicitly generic categories that fit no family and are deliberately retained in the served “other” bucket). Families are *operationally analogous planning buckets*, not identical workloads: Chicago has no noise family at all (residential noise is not a Chicago 311 request type), the supplement reports the full native-category composition of every family in every city, and no cross-city claim in this paper assumes compositional identity.

Weather. NOAA GHCN-Daily station observations (precipitation, max/min temperature, snow-fall, snow depth; derived mean temperature) from one first-order station per city (Central Park; O’Hare; San Francisco Downtown; Camp Mabry), retrieved from NOAA’s AWS Open Data mirror with checksums. Quality-flagged values are dropped; short temperature gaps (≤ 7 days) are interpolated; missing snow fields are treated as zero. Wind speed was excluded for all cities because the San Francisco station does not report it and Central Park has a multi-month gap — a documented deviation from the single-city study this work extends. A single station per city is a documented spatial simplification (A2). No synthetic demand or synthetic weather is used anywhere in the study.

5 Methods

Feature sets. Four nested feature sets isolate the marginal value of each public signal: *internal* (lags 1, 2, 3, 7, 14, 28 and 7/28-day rolling mean and standard deviation, windows ending at day t); *calendar* (internal + deterministic day-of-week, month, weekend, U.S. federal holiday, and smooth day-of-year terms for the target day); *weather* (internal + target-day and current-day station weather); and *calendar+weather*. Family indicators are one-hot encoded; pooled models add city indicators.

Models. Two forecast-free baselines (trailing 7-day mean; same-weekday seasonal naive), ridge regression, a random forest (Breiman, 2001), and LightGBM with an L1 objective (Ke et al., 2017) form the pre-specified point-model grid; quantile-objective LightGBM (following Koenker and Bassett, 1978) provides the predictive distribution, with non-crossing enforced by cumulative maxima. Hyperparameters are fixed across cities and feature sets; baselines are never dropped from tables.

Protocol. Within each city, the earliest 70% of days form training, the next 15% validation, and the final 15% an untouched test window. For every (city, feature set), the model is selected on VALIDATION MAE, frozen, refit on train+validation, and evaluated once on test; exhaustive test

grids appear only as supplementary descriptive evidence. The quantile model is fit on training data alone for all validation-stage outputs and refit on train+validation for its single test evaluation. Stability is assessed with five expanding-window rolling-origin folds (Hyndman and Athanasopoulos, 2021). Pooled (“global”) models train on all four cities with stage-specific temporal censoring: validation-stage fits use only rows dated no later than the earliest training cutoff across cities, test-stage fits no later than the earliest validation cutoff, so no pooled training row is contemporaneous with any evaluation window. Leave-one-city-out transfer is likewise censored at the held-out city’s validation cutoff. Two selection procedures with explicitly different candidate sets are used and named throughout: *within-scope selection* (per city, feature set, and scope, used for every headline table and figure) selects among the five point models of that scope only; the *cross-scope selection experiment* (Section 7) selects over a pre-specified eleven-configuration grid that pools the ten local configurations with the stage-censored global calendar+weather model. All accuracy contrasts use paired moving-block bootstrap intervals on daily mean absolute-error differentials (28-day blocks, 2,000 resamples, seed 20260609; 14- and 56-day blocks as pre-specified sensitivity) (in the spirit of Diebold and Mariano, 1995).

Decision evaluation. Every point forecaster drives the proportional and greedy policies; the quantile model drives the three uncertainty arms; uniform allocation is the forecast-free floor. Simulations run day-by-day over each city’s test window at the frozen budgets with $\kappa = 50$ and equal weights. Pre-specified sensitivities: one normative-priority scenario; one-family-at-a-time service-yield multipliers of 0.70 and 1.30 over every active family; 10% abandonment; train+validation budget calibration; and an Austin variant that excludes the heterogeneous *other* family with budgets recomputed accordingly. Validation-time model selection by decision loss (simulated on the validation window at the same frozen budgets) is compared against selection by validation MAE, both frozen before test.

Guard suite. Thirteen automated guards enforce the protocol and fail the build otherwise, including: budgets reproducible from training data and invariant to test perturbation; selection invariant to test metrics; recomputed censoring dates on every transfer row and a fit-time proof manifest for pooled censoring; identical fitted model behind all uncertainty arms; structural absence never encoded as zero; exact budget conservation; forbidden-terminology scans of all manuscript-facing artifacts; configuration–output agreement for every sensitivity scenario; source-identifier consistency; and provenance hashes binding every table and figure to the generating run.

6 Forecasting Results

All headline aggregations use validation-only selection; exhaustive grids are supplementary. Intervals are paired 28-day moving-block bootstrap (stable at 14 and 56 days throughout).

Calendar gains replicate everywhere. The validation-selected calendar model lowers held-out test MAE relative to the validation-selected internal-history model in every city: -11.9% in New York ($207.0 \rightarrow 182.4$), -15.6% in Chicago ($60.4 \rightarrow 50.9$), -10.4% in San Francisco ($35.2 \rightarrow 31.5$), and -10.2% in Austin ($16.5 \rightarrow 14.8$); the fixed-estimator (LightGBM) calendar effect is significant in all four cities. Against the same-weekday seasonal-naïve baseline, the selected calendar+weather model reduces test MAE by 41.4% (New York), 41.5% (Chicago), 31.3% (San Francisco), and 31.3% (Austin). Validation-only selection matters in its own right: on the internal feature set it picks the

random forest in Austin and Chicago and ridge in San Francisco, choices a test-minimum rule would have hidden.

Weather’s marginal value is real but city-dependent. On top of calendar features, target-day weather further reduces MAE by 9.3% in New York, 3.3% in Chicago, and 4.7% in San Francisco (all intervals excluding zero) — and by exactly nothing in Austin (+0.001 MAE; interval covers zero). The Austin null is retained as a finding. The lagged-weather variant (assumption A3 removed) erases most of New York’s weather gain (178.6 versus 165.4, against a 182.4 calendar-only reference), so much of the weather value is conditional on an accurate day-ahead weather forecast — a realistic but explicitly stated premise.

Stability. Five expanding-window rolling-origin folds reproduce the feature-set ordering (internal > calendar > calendar+weather in MAE) in every city; the headline comparisons are not artifacts of a single split.

Pooling and transfer mostly hurt under valid censoring. With stage-censored training (no pooled training row contemporaneous with any evaluation window), the pooled model raises test MAE relative to local models in Austin (+4.1%), Chicago (+6.5%), and New York (+1.8%), each interval excluding zero, and is neutral in San Francisco (−1.2%, interval covers zero). Two distinct selection procedures operate in this paper and must not be conflated: the *within-scope* headline selection (Section 5) chooses among local candidates for the local headlines, where the local LightGBM wins in every city; the *cross-scope* selection experiment of Section 7 chooses over the full eleven-configuration grid, where validation MAE still prefers the pooled model in Austin (validation MAE 15.08 vs. 15.25 local, despite worse test error) and in San Francisco (27.20 vs. 28.97, where the pooled model is also genuinely better on test). Temporally censored zero-shot transfer (training without the target city, censored at its validation cutoff) degrades MAE by +5.4% (San Francisco) to +148.5% (New York). With six years of local history, cross-city pooling and transfer are not beneficial in this benchmark; the negative result is retained.

Probabilistic forecasts are sharp but under-dispersed. The quantile model’s central 90% intervals cover 76.8–81.2% of test outcomes. We report this miscalibration rather than re-tuning it away; Section 7 shows the distributional information is nonetheless valuable within the optimization policy class, and conformal recalibration is noted as future work.

7 Simulated Capacity-Allocation Evaluation

All simulations run at budgets frozen from training data (New York 119/153/187 units across the scarce/moderate/generous regimes; Chicago 29/38/46; San Francisco 26/33/41; Austin 8/11/13; $\kappa = 50$), under the neutral equal-weights objective. Losses are waiting-time-weighted unresolved stocks (Section 3); policy contrasts carry paired 28-day block-bootstrap intervals on the daily loss differentials. Because the trend diagnostic of Section 8 finds 45 of the 48 differential series trend-dominated (the carryover stock compounds policy differences), we read these intervals as *descriptive characterizations of a systematic, partly deterministic divergence within the simulation*, retaining a stationary-inference reading only where the diagnostic supports it (Chicago, generous regime). The simulation itself is deterministic given the frozen inputs, so every directional statement below is exact for this environment; what the trending differentials preclude is generalizing the interval widths as stationary sampling uncertainty. Four findings emerge.

Point forecasts disable the expected-value optimizer; a simple proportional rule is competitive with even its distribution-aware form. Two separate statements, with separate strength. First, feeding *point* forecasts into the expected-value optimizer is a failure mode: it loses to naive proportional apportionment in all 12 city-regime settings. The mechanism is identified and general: under a degenerate forecast distribution the optimizer’s marginal-value profile is piecewise flat — equal to $\kappa\omega$ for every unit below the point estimate plus carryover, and zero for every unit above it — so under a neutral objective the greedy ordering carries no information in either the deficit or the surplus segment, and allocation is decided by tie-breaking (worked examples for both segments are in the supplement). Second, against the *fair* comparator — the full-distribution arm of the same quantile model, whose tails restore an allocation gradient — proportional apportionment remains competitive to slightly better: lower loss in 11 of 12 settings by 0.1–6.5%, with one reversal (San Francisco, generous). This residual edge is an observed pattern for which we do not claim a demonstrated mechanism; a plausible candidate — per-day-myopic optimality is not horizon optimality under persistent carryover, and proportional apportionment approximates a smoother hedge — is left as a conjecture. Relative to the forecast-free uniform floor, the proportional policy with the selected forecaster reduces the cumulative unresolved stock by 18–23% (scarce), 38–70% (moderate), and 60–93% (generous), the regime pattern itself being partly mechanical: under structural under-capacity no policy can prevent stock growth, so policy differences compress. All of this is a property of the simulated environment and its neutral objective — under the normative-priority sensitivity the gap compresses — and constitutes no guidance about any city’s actual operations.

Within the optimization family, the full distribution beats its own point representations everywhere. Holding the single fitted quantile model and every other input fixed, the full-distribution arm outperforms both the median arm and the implied-mean arm in 12 of 12 settings, with all paired intervals excluding zero. Distributional information breaks the flat-profile ties that disable the point-forecast optimizer — evidence that *how* a forecast enters a policy can matter as much as forecast accuracy itself. One caveat is owed: the distribution whose value is measured is itself imperfect — its 90% intervals cover only 77–81% of outcomes (Section 6) — so this result demonstrates the value of the *available, under-dispersed* predictive distribution in this simulation, not of calibrated uncertainty; it is plausibly a lower bound on the latter.

Forecast quality transfers to simulated decision value within a fixed policy. Replacing the trailing-mean baseline with the validation-selected forecaster, policy held fixed, lowers the cumulative unresolved stock in all 12 settings (average daily reduction in unresolved *stock* from ≈ 1.8 thousand request-equivalents in Austin to ≈ 72 thousand in New York under scarcity). Consistently, model *rankings* by test MAE and by simulated decision loss agree closely across the eleven-configuration grid (Spearman $\rho = 0.88$ – 0.99 in every city-regime cell): in this environment, accuracy is largely a sufficient proxy for simulated decision value among reasonable models.

Accuracy-based and decision-based selection mostly agree. Selecting over the cross-scope eleven-configuration grid by validation decision loss instead of validation MAE — both under the frozen budgets — yields the same choice in 9 of 12 settings, a gain in 2 (Austin, scarce and moderate: validation MAE prefers the pooled model there, the decision criterion prefers the local one, and the local choice is better on the held-out test), and one interval-tie; it never harmed. In San Francisco both criteria select the pooled model, which is also the better test model there. Decision-based validation selection is therefore a cheap, safe check rather than an established improvement.

8 Robustness and Sensitivity

Trend diagnostic and inference framing. For every city, regime, and pre-named contrast we regress the daily loss differential on the day index and test the slope with a Newey–West (lag-28) statistic, reporting also the share of the mean differential attributable to the linear drift (supplement, Table tab10). The result is unambiguous: 45 of 48 differential series are trend-dominated (the only stationarity-compatible cell is Chicago under generous capacity), and 11 of 12 loss *level* series trend as well. This is a structural property of the carryover stock, strongest under scarcity but present broadly: policy differences compound rather than fluctuate around a stationary mean. Accordingly, all decision-contrast intervals in Section 7 are read as descriptive characterizations of a systematic divergence within the deterministic simulation; a stationary-inference reading is retained only for the Chicago/generous cell. The forecast-accuracy contrasts of Section 6 are unaffected — their daily error differentials carry no stock — and their inferential reading stands.

Block-length stability. None of the 48 decision-contrast interval conclusions changes when the bootstrap block length moves from the primary 28 days to 14 or 56 days; the same holds for the forecast contrasts. (Block-length stability does not, by itself, detect a common trend — which is why the explicit trend diagnostic above is the binding check.)

Decision-layer sensitivity. The pre-specified grid preserves every qualitative conclusion: with 10% daily abandonment, policy ordering is unchanged in all cities; with one-family-at-a-time service-yield multipliers of 0.70 and 1.30 over every active family, the ordering direction never reverses (the largest perturbation is New York, generous regime, noise yield $\times 1.30$); with budgets calibrated on train+validation instead of training only, conclusions are unchanged; and the Austin analysis with the heterogeneous *other* family excluded (budgets recomputed) produces the identical policy ordering as the primary analysis, so Austin’s conclusions do not depend on its residual category. Under the illustrative normative-priority scenario the proportional-optimizer gap compresses, which is why the proportional-dominance claim is stated as conditional on the neutral objective.

What would change the conclusions. The simulation results are conditional on the stylization assumptions (interchangeable units, fixed yield, no intra-day dynamics) and on the equal-weights objective; we claim regime *patterns* within this simulated environment, not city-specific magnitudes. The forecasting results are unconditional on those assumptions; the weather findings are conditional on day-ahead forecast availability (A3), bounded by the lagged-weather variant.

9 Responsible Use of Reported-Demand Data

311 records measure the propensity to report as much as the incidence of problems: communities differ systematically in access, trust, language, and expectation that reporting leads to action (Kontokosta et al., 2017; Kontokosta and Hong, 2021), and prediction-driven prioritization can institutionalize whatever bias the data carry (Samorani et al., 2022). Three design consequences follow in this study. First, the target variable is named throughout as *reported* demand, and no result is interpreted as a measurement of need. Second, because our spatial unit is the whole city, the most direct channel for reporting-bias amplification — reallocating resources *away* from low-reporting neighbourhoods — is outside the action space of the simulated planner; the family-level channel remains, and we report per-family unserved-demand distributions for every policy so that systematic deprioritization of any family is visible rather than hidden in aggregate scores. Third, the priority

weights ω_s encode a value judgment, not a fact; we expose them as configuration, vary them over a pre-specified grid in sensitivity analysis, and regard their selection in any real deployment as a decision that belongs to accountable public officials, not to a model. A forecast-driven allocation system used in practice would additionally require visibility into where reporting data are likely incomplete — a measurement problem this paper’s data cannot solve and that we therefore flag as a limitation rather than simulate.

10 Limitations

- **Explicit simulation boundary.** The allocation layer is a transparent simulation over abstract request-equivalent capacity units; it omits shifts, travel, specialization, geography, and intra-day dynamics, and none of its parameters estimates any city’s actual capacity, productivity, or queues. Conclusions concern the structure of the accuracy-to-allocation mapping inside this simulated environment.
- **Queue instability and horizon dependence.** Under the scarce (and to a lesser degree moderate) regimes the simulated system is structurally under-capacitated, so the unresolved stock grows over the evaluation horizon; cumulative-loss magnitudes and percentage reductions therefore depend on the horizon length, regime comparisons are partly mechanical (differences compress under supercriticality), and decision-contrast intervals are descriptive rather than stationary inference (Section 8).
- **Objective dependence.** The proportional-dominance result is conditional on the neutral equal-weights objective; the normative-weight sensitivity compresses the gap. No weighting here is socially validated.
- **Reported demand only.** Targets and losses are functions of reported requests; unreported need is invisible to both the forecasters and the simulated planner (Section 9).
- **Harmonization is a modeling choice.** The service families are deterministic keyword buckets; native taxonomies differ across cities (Chicago lacks a residential-noise intake type; Austin’s residual *other* is large, though its exclusion leaves conclusions unchanged). All mappings and compositions are published.
- **Weather information set.** Target-day weather uses realized observations as a proxy for a day-ahead forecast (A3), bounded by the lagged-weather variant; one station per city coarsens spatial variation (A2); wind was excluded for data-completeness reasons.
- **Calibration.** The quantile model’s intervals are under-dispersed (90% nominal, 77–81% empirical); conformal recalibration is future work.
- **Scope.** Four U.S. Socrata-based systems; fixed a priori hyperparameters; no deep or foundation models; no causal claims; no deployment; no measurement of social need.

11 Conclusion

We built a guarded, reproducible four-city benchmark of next-day reported 311 demand and coupled it to a controlled, explicitly simulated capacity-allocation evaluation with frozen pre-test environments and validation-only selection. The corrected evidence supports a precise set of claims: public calendar signals improve next-day forecasts in every system and weather helps heterogeneously;

temporally valid pooling and transfer mostly hurt; within the simulation, *how* a forecast enters the allocation rule can matter as much as which forecaster produced it — point-forecast inputs disable an expected-value optimizer entirely, the full predictive distribution of a single quantile model restores its gradient in every setting, and a simple proportional rule remains competitive with even the distribution-aware optimizer under the tested regimes and neutral objective; and decision-aware validation selection mostly coincides with accuracy-based selection while never harming. The broader lesson for forecast-driven public operations research is methodological: evaluation environments must be frozen before selection, selection must never touch test outcomes, and the interface between a forecast and a policy deserves the same scrutiny as the forecast itself. Natural continuations include conformal recalibration, richer service-time modeling where closure data permit, spatial disaggregation where geographies allow honest comparison, and extension beyond U.S. systems as comparable public data emerge.

Reproducibility statement. All data are public; the companion repository contains acquisition scripts with exact query URLs, SHA-256 manifests for every raw file, deterministic seeds, unit tests (including leakage and allocation-optimality checks), and a **Makefile** that regenerates every number, table, and figure in this paper from the versioned raw data with a single command.

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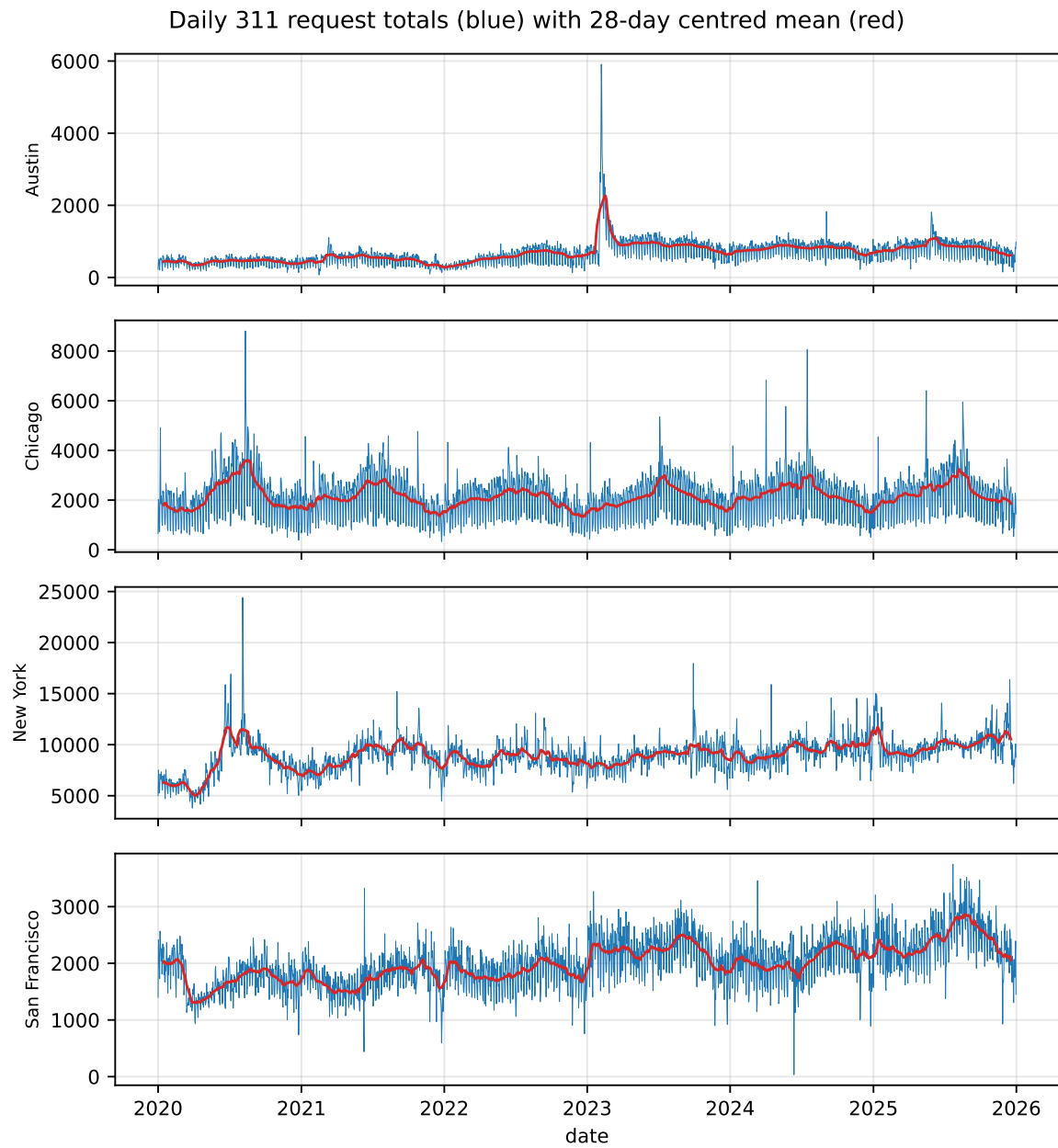


Figure 1: Daily citywide 311 request totals (blue) with 28-day centred mean (red), 2020–2025. Scale, seasonality, and disruption patterns differ visibly across the four systems.

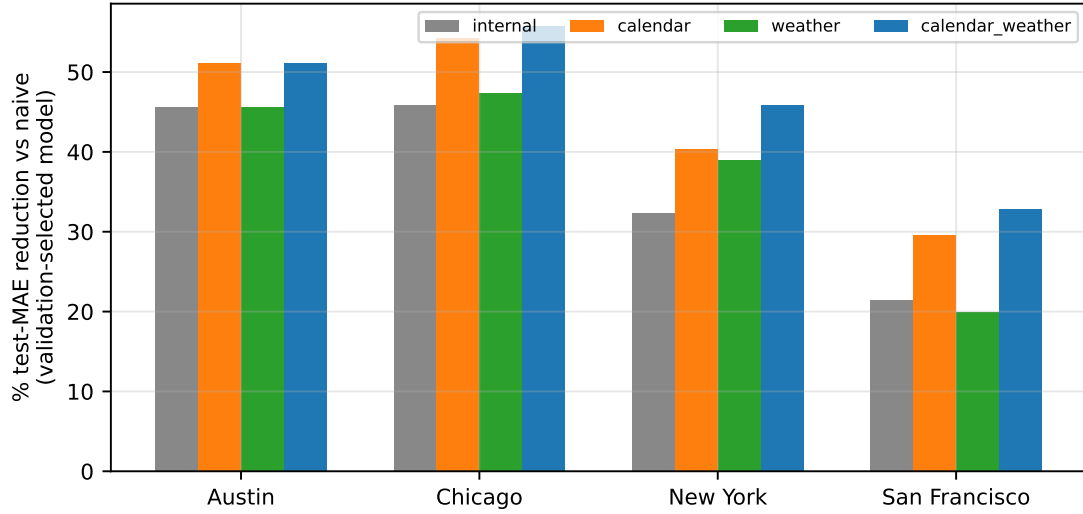


Figure 2: Held-out test-MAE reduction versus the trailing-mean baseline for the validation-selected model of each feature set and city.

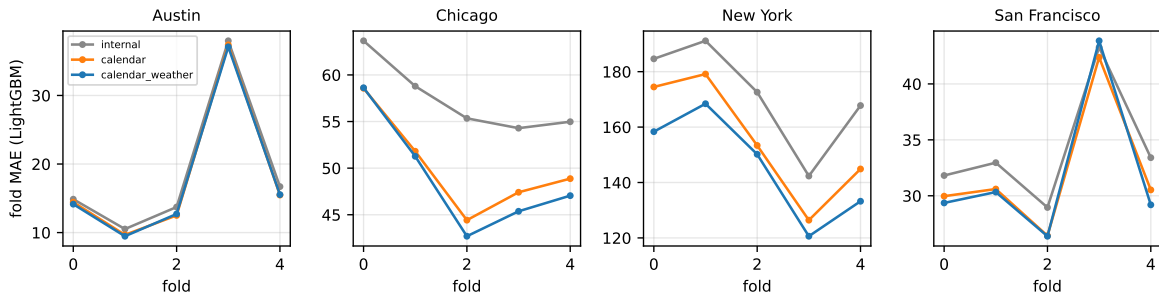


Figure 3: Rolling-origin fold MAE (LightGBM) by feature set: the calendar and weather orderings are stable across folds in all cities.

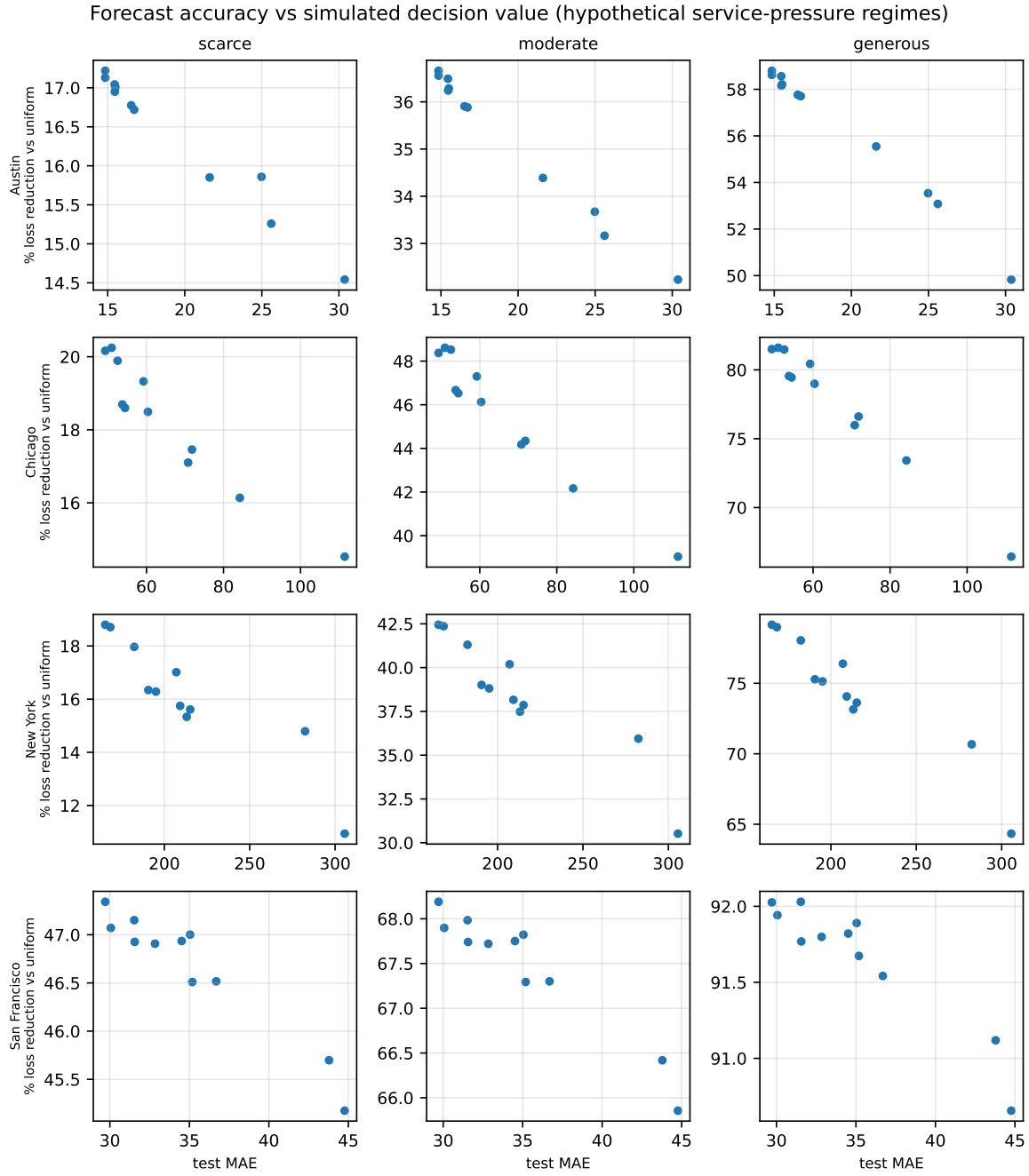


Figure 4: Forecast accuracy (test MAE, x) versus simulated decision value (% loss reduction relative to the uniform floor, y) for the pre-specified point-forecast grid under the greedy expected-value policy, by city and hypothetical service-pressure regime.